

Module 8:

Mixture Models, EM, and Variational Inference

A single merged treatment of Bishop Ch. 9 and §10.1–10.3 and ESL §8.5,
standing on Foundations B §B.8 and Foundations D

Prepared for Tom Koreslesjak · companion to `NEC_thesis_syllabus.md`

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Abstract

The conceptual core of the thesis, and the chapter the whole book has been arranging furniture for. The Gaussian mixture model is developed in full — both of its log-likelihoods, EM executed with closed-form M-steps, and the three failure modes (collapse, label switching, local optima) traced to structural properties of the likelihood rather than recited as folklore. The variational Bayesian mixture is treated structurally: what priors buy (no singularities, automatic component pruning) and what algebra is deferred. The chapter then performs the translation that matters most here: GMM to mixture of experts — same E-step, M-steps that are precisely Module 4’s weighted problems, and a proof-backed identification of SGD-on-the-NLL with generalized EM. By the end, the corrected NEC’s training loop is not a recipe but a theorem with an implementation.

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How this chapter was assembled (and what was cut)

This module inherits more finished work than any other, and the de-duplication ledger is correspondingly long. *Already proven, cited not repeated*: EM’s monotonicity (Foundations D, Ex. D.3(d)); EM as coordinate ascent on the ELBO (D, eq. D.5.4 and the capstone); the log-sum-exp trick (B, eq. B.6.10 — the syllabus re-assigns its derivation here, discharged by citation, with the genuinely new part — the log-domain mixture pipeline — kept as Exercise 8.1); the responsibility formula from Bayes (B.8.4, D §D.8); mixtures generate fat tails (B, Ex. B.6). *Assigned problems that duplicate finished work*: Bishop 10.1 is Exercise D.3(a) verbatim — folded, not re-assigned; ESL 8.1 and 8.2 are Gibbs’ inequality (B, Ex. B.5) and the Gibbs-maximizer result (D, Ex. D.3(c)) in ESL’s notation — converted into the notation-bridge Exercise 8.9, which is the honest version of assigning them. *Choices within assignments*: of Bishop 10.5/10.10/10.12 the syllabus asks for one; 10.5 is chosen — it proves that variational inference with a point-mass over parameters is EM, the unification theorem this book’s architecture deserves — with 10.12 (the full conjugate VB-GMM verification) flagged as optional for the dedicated. *Cuts*: Bishop §9.1’s K -means (kept as one paragraph: initialization device and hard-EM limit); Bishop §10.2’s conjugate algebra (Dirichlet, Normal–Wishart, digamma corrections — the chapter presents the *structure* and the two payoffs, defers the algebra to Cookbook §8 plus Bishop’s text, consistent with the conjugacy cut standing since Foundations B); Bishop §10.3 (VB linear regression) and §10.4+ (cut entirely); ESL §8.5’s Gibbs-sampling segue (one sentence).

Notation

Data $x_1, \dots, x_N \in \mathbb{R}^p$ (for the unconditional mixture) or pairs (x_n, y_n) (for the mixture of experts). Latent labels $z_n \in \{1, \dots, K\}$, one-hot when convenient. Responsibilities $\gamma_{nk} = p(z_n = k \mid \text{data}_n, \theta)$, component counts $N_k = \sum_n \gamma_{nk}$. Mixture weights π on the simplex. All cross-references follow the ledger: (B.·), (C.·), (D.·), (4.·)–(7.·).

8.1 Latent-variable models and the Gaussian mixture

A *latent-variable model* explains observed data as the visible half of a joint distribution $p(x, z)$ whose hidden half z encodes structure — here, *which regime generated this observation*. The Gaussian mixture model is the canonical instance:

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k), \quad \pi \in \Delta^{K-1}, \quad (8.1.1)$$

generatively: draw $z \sim \text{Cat}(\pi)$, then $x \sim \mathcal{N}(\mu_z, \Sigma_z)$ — the sum rule marginalizing z out, exactly the construction of (B.8.1) with the gate constant and the “expert” a density over x itself rather than over $y \mid x$. Keep that dictionary in view from the start: *the GMM is the MoE with the inputs deleted*; everything derived in this chapter transfers to the thesis model by reinstating input-dependence at marked spots (§8.6).

Two structural facts before any fitting. *Moments*: the mixture’s mean and covariance are $\mathbb{E}[x] = \sum_k \pi_k \mu_k$ and $\text{Cov}(x) = \sum_k \pi_k (\Sigma_k + \mu_k \mu_k^\top) - \mathbb{E}[x] \mathbb{E}[x]^\top$ (Bishop’s 9.49–9.50; Exercise 8.6 derives them by the tower property and adds the fourth-moment punchline you already own from B, Ex. B.6: scale mixtures are leptokurtic — the statistical argument *for* regime structure in returns). *Identifiability*: the likelihood is invariant under the $K!$ relabelings of components — the same permutation symmetry that made neural losses non-convex (Module 6 §6.1) — so “component 2” has no meaning across runs, folds, or random seeds until you impose an ordering convention. This is not pedantry; it returns as a thesis-reporting requirement in §8.4.

Finance reality check

Why mixtures, for returns, in one paragraph you can give a committee: Foundations B §B.7.3 proved that mixing Gaussians with unequal variances produces excess kurtosis — the unconditional fat tails of returns are *exactly* what conditionally-Gaussian regimes look like after marginalization. A GMM fit to returns is therefore not a curve-fitting convenience; it is the minimal model in which “calm market, stressed market” is a hypothesis with a likelihood. The thesis’s MoE upgrades the hypothesis from *states exist* to *states are predictable from features* — and this chapter builds the estimation theory for both.

8.2 Two log-likelihoods, one gap

Everything about EM follows from comparing two objectives. The *incomplete-data* log-likelihood — the one you actually want to maximize — is

$$\ell(\theta) = \sum_{n=1}^N \log p(x_n | \theta) = \sum_{n=1}^N \log \sum_{k=1}^K \pi_k \mathcal{N}(x_n | \mu_k, \Sigma_k) = \sum_{n=1}^N \text{LSE}_k(\log \pi_k + \log \mathcal{N}(x_n | \mu_k, \Sigma_k)), \quad (8.2.1)$$

a sum of log-sum-exps (B §B.6.3 — compute it in the log domain or watch it underflow; Exercise 8.1 quantifies exactly when). The log outside the sum couples all components: no closed-form maximizer exists, gradients exist but the objective is non-convex with the symmetries of §8.1.

The *complete-data* log-likelihood — the one you would maximize if the labels were visible — is, with one-hot z_{nk} ,

$$\log p(X, Z | \theta) = \sum_{n=1}^N \sum_{k=1}^K z_{nk} [\log \pi_k + \log \mathcal{N}(x_n | \mu_k, \Sigma_k)], \quad (8.2.2)$$

and it is a *pleasure*: the log sits inside, the objective separates by component, and maximization is Foundations B-grade statistics — each Gaussian fitted to its own points, mixing weights equal to label fractions (Exercise 8.7, Bishop 9.7). The entire difficulty of mixture estimation is the gap between (8.2.1) and (8.2.2): the labels are missing. EM’s resolution, prefabricated in Foundations D: *replace the missing labels by their posterior expectations and iterate* — maximize $\mathbb{E}_q[\log p(X, Z | \theta)]$, which keeps (8.2.2)’s separability while honestly averaging over label uncertainty, and choose q to make the surrogate tight.

Checkpoint exercise

Exercise 8.1 [Syllabus, de-duplicated: the log-domain mixture pipeline]. [★★★★ Core] *The LSE identity and max-shift stabilization are (B.6.10); cite, do not re-derive. The new, mixture-specific content: (a) Show that the naive pipeline — evaluate each $\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ as a number, multiply by π_k , sum, take log — underflows in float64 whenever every component’s squared Mahalanobis distance exceeds $\approx 2 \times 745 \approx 1490$ (a Mahalanobis radius near 39σ) (use the float64 underflow threshold $\approx e^{-745}$ and the Gaussian log-density’s form); convert that to a Mahalanobis radius and note how routinely a fat-tailed return panel (B §B.7) exceeds it. (b) Write the correct pipeline: $a_{nk} = \log \pi_k + \log \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ computed directly in the log domain (no exponentials anywhere), $\ell_n = \text{LSE}_k(a_{nk})$ via max-shift, $\gamma_{nk} = \text{softmax}_k(a_{nk})$ (Module 4’s fused machinery). Verify the responsibilities are exact even when every $e^{a_{nk}}$ would underflow. (c) One sentence: which earlier chapter’s “library fused-op” lesson is this the mixture version of?*

8.3 EM for the Gaussian mixture, executed

Foundations D supplied the theory whole: coordinate ascent on $\mathcal{L}(q, \theta)$, E-step closing the KL gap, M-step raising the floor, likelihood monotone (D eq. D.5.4; monotonicity proven as Ex. D.3(d)). Here is its execution on (8.1.1).

E-step. The optimal q is the exact posterior — for a discrete label, a K -vector per sample, computable exactly (no variational approximation needed at this stage; D §D.8 made the point): the *responsibilities*

$$\gamma_{nk} = \frac{\pi_k \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)} = \text{softmax}_k(\log \pi_k + \log \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)), \quad (8.3.1)$$

Bayes’ rule as promised since (B.8.4), computed in the log domain via the softmax of log-joints (Module 4’s machinery, Exercise 8.1’s pipeline).

M-step. Maximize the expected complete-data log-likelihood $Q(\theta) = \sum_{n,k} \gamma_{nk} [\log \pi_k + \log \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)]$ — (8.2.2) with soft labels. The closed forms, whose from-scratch derivation is the syllabus’s centerpiece Exercise 8.2:

$$\boldsymbol{\mu}_k = \frac{1}{N_k} \sum_n \gamma_{nk} \mathbf{x}_n, \quad \boldsymbol{\Sigma}_k = \frac{1}{N_k} \sum_n \gamma_{nk} (\mathbf{x}_n - \boldsymbol{\mu}_k)(\mathbf{x}_n - \boldsymbol{\mu}_k)^\top, \quad \pi_k = \frac{N_k}{N}, \quad (8.3.2)$$

with $N_k = \sum_n \gamma_{nk}$ the *effective count* claimed by component k . Every line is an old friend wearing soft weights: the mean is a responsibility-weighted average — optimal, not just natural, by the Bregman result (D, Prop. D.3.1); the covariance is the weighted MLE of (B §B.4.2), Bessel caveat included in spirit; the weights are label fractions, derived under the simplex constraint by exactly the KKT machinery of Module 4 §4.3; and the matrix calculus that produces $\boldsymbol{\Sigma}_k$ runs on the $\partial \log |\cdot|$ identity planted in Foundations C §C.6 for precisely this moment. Iterate (8.3.1)–(8.3.2) to a fixed point: each sweep costs $O(NKp^2)$, the likelihood ascends monotonically, and convergence is to a stationary point — about which the next section is the honesty.

Remark 8.3.1 (K -means, in passing). Freeze every $\boldsymbol{\Sigma}_k = \epsilon \mathbf{I}$ and let $\epsilon \rightarrow 0$: responsibilities harden to nearest-centroid indicators and (8.3.2) becomes the K -means update. K -means is hard-assignment

EM on a degenerate GMM — which is why it is the standard *initializer* for EM (cheap, robust), and why its failure modes (sensitivity to scaling, spherical bias) are the degenerate cases of this chapter’s.

Checkpoint exercise

Exercise 8.2 [Syllabus centerpiece: the M-step from scratch]. [★★★★ Core] Maximize $Q(\theta) = \sum_{n,k} \gamma_{nk} [\log \pi_k + \log \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)]$ with responsibilities frozen. (a) Means: set $\nabla_{\boldsymbol{\mu}_k} Q = 0$ (the quadratic-form gradient of Foundations A) and derive $\boldsymbol{\mu}_k = N_k^{-1} \sum_n \gamma_{nk} \mathbf{x}_n$; confirm via D, Prop. D.3.1, that this weighted mean is the unique Bregman-optimal center, not merely a stationary point. (b) Covariances: derive the $\boldsymbol{\Sigma}_k$ update of (8.3.2). (Route: parameterize by the precision $\boldsymbol{\Lambda}_k = \boldsymbol{\Sigma}_k^{-1}$; use $\nabla_{\boldsymbol{\Lambda}} \log |\boldsymbol{\Lambda}| = \boldsymbol{\Lambda}^{-\top}$ — the identity stated for this purpose in Foundations C §C.6 — and the linearity of the quadratic form in $\boldsymbol{\Lambda}$ via the trace trick (B.1.10’s proof pattern); solve and invert.) (c) Weights: maximize over $\boldsymbol{\pi}$ on the simplex with a Lagrange multiplier (KKT, Module 4 §4.3) and obtain $\pi_k = N_k/N$; identify the multiplier’s value and its shadow-price meaning. (d) Verify second-order/boundary sanity: why is the $\boldsymbol{\Sigma}_k$ stationary point a maximizer for $N_k > p$, and what happens to it as $N_k \rightarrow 0$ (connect to §8.4, mode 1)?

8.4 How mixture estimation fails, structurally

Three failure modes, each traceable to a structural property of (8.2.1); Exercise 8.3 works all three.

1. *Collapse: the unbounded likelihood.* Center one component on a single data point and shrink its variance: with $\boldsymbol{\mu}_k = \mathbf{x}_1$ and $\boldsymbol{\Sigma}_k = \sigma^2 \mathbf{I}$, that component contributes density $\propto \sigma^{-p} \rightarrow \infty$ while every other point remains covered by the remaining components — so $\sup_{\theta} \ell(\theta) = +\infty$ and the GMM has no maximum-likelihood estimator at all. EM does not always find these singular spikes, but they are always there, adjacent to good solutions, and small- N_k components drift toward them. The character of the failure is worth naming: it is the continuous density’s fault (a density may exceed 1; B §B.1.1), and discrete-emission mixtures are immune — a Bernoulli mixture’s likelihood is bounded above (Exercise 8.8(b) proves it, the instructive contrast). Remedies, in increasing principle: variance floors ($\boldsymbol{\Sigma}_k \succeq \epsilon \mathbf{I}$ enforced); MAP-EM with a prior on $\boldsymbol{\Sigma}_k$ (Module 4 §4.5’s move — the M-step gains a shrinkage term); or the Bayesian treatment of §8.5, where the singularity vanishes outright.

2. *Label switching.* The $K!$ symmetry of §8.1 means the likelihood surface has $K!$ copies of every optimum. Harmless for prediction (the mixture density is invariant), poisonous for *interpretation*: “expert 2 is the stress regime” is meaningless across restarts, folds, or bootstrap replicates unless components are matched — by sorting on a statistic (e.g. component variance), or by greedy alignment to a reference run. Any thesis table that averages per-regime quantities across folds without an alignment rule is averaging apples with relabeled oranges.

3. *Local optima.* Non-convexity here is not the benign deep-learning kind: distinct local optima correspond to genuinely different *clusterings*, often with materially different likelihoods. Practice: initialize from K -means (or several), run multiple seeds, keep the best held-out log-likelihood — and log every restart in the Module 5 trial registry, because “best of 20 initializations” is a selection

event like any other.

NEC connection

Each failure mode has a gate-side twin that the corrected NEC inherits: collapse $\leftrightarrow$ a dead expert whose responsibilities (hence gradients, B.8.5) vanish — the utilization diagnostics and load-balancing devices of (B §B.6’s box) are the MoE’s variance floor; label switching $\leftrightarrow$ “regime k ” must be aligned across walk-forward folds before any per-regime IC table is averaged (Module 5’s protocol gains a clause here); local optima $\leftrightarrow$ initialization of the gate near-uniform with *diverse* experts, never the reverse (a confident random gate at $t = 0$ is a collapsed mixture before training starts — §8.7 operationalizes).

Checkpoint exercise

Exercise 8.3 [Syllabus: the three failure modes, constructed]. [★★★★ Core] (a) Collapse. For a two-component GMM in one dimension with $\mu_1 = x_1$ and $\sigma_1 = \epsilon$, lower-bound the likelihood by a function diverging as $\epsilon \rightarrow 0$ (keep component 2 fixed and covering the rest), establishing $\sup_{\theta} \ell = \infty$; identify the divergence rate in ϵ and explain why EM iterations approach such spikes only when some $N_k \rightarrow \text{small}$ (connect to (8.3.2)’s covariance with one point). (b) Label switching. Show formally that the likelihood is invariant under any permutation $\tau \in S_K$ applied jointly to (π_k, μ_k, Σ_k) ; conclude every optimum has $K!$ copies; then write the two-line protocol clause the thesis needs: an alignment rule for “regime k ” across walk-forward folds, and what statistic you would sort on for volatility regimes. (c) Local optima. Construct (describe precisely; no computation needed) a one-dimensional dataset and two initializations for $K = 2$ that converge to materially different fitted mixtures, and state the practice that Module 5’s registry imposes on “best of m restarts.”

8.5 Variational Bayesian mixtures, structurally

Bishop §10.2 promotes every GMM parameter to a random variable — Dirichlet prior on π , Gaussian–Wishart priors on (μ_k, Σ_k^{-1}) — and approximates the joint posterior over labels *and* parameters by the mean-field factorization of Foundations D §D.6:

$$q(\mathbf{Z}, \pi, \mu, \Sigma) = q(\mathbf{Z}) q(\pi, \mu, \Sigma), \quad (8.5.1)$$

cycled through the cavity update (D, eq. D.6.2). This chapter keeps the treatment *structural* — the factorization, the update cycle’s shape, and the two payoffs — and defers the conjugate algebra (digamma functions, Wishart expectations, Cookbook §8’s Gaussian moments) to Bishop’s text, consistent with the conjugacy cut standing since Foundations B; Exercise 8.4 (Bishop 10.5) proves the structural theorem that earns the section its place, and Bishop 10.12 is flagged, optional, for whoever wants the full verification.

The update cycle’s shape: $q(\mathbf{Z})$ ’s update looks exactly like the E-step (8.3.1) except that $\log \pi_k$ and $\log \mathcal{N}$ are replaced by their *expectations under the parameter posterior* — EM with averaged parameters; $q(\pi, \mu, \Sigma)$ ’s update looks exactly like the M-step (8.3.2) except that counts and

moments update *prior pseudo-counts* instead of overwriting them — the M-step with Bayesian shock absorbers. Two payoffs justify the machinery:

No singularities. The Wishart prior puts zero mass on degenerate covariances; the offending $\sigma \rightarrow 0$ spike of §8.4 is integrated away rather than patched — the principled version of the variance floor, and the cleanest available answer to the GMM’s ill-posed MLE.

Automatic pruning. Components the data do not need have their effective counts driven toward zero, their parameter posteriors collapse back to the prior, and their expected mixing weights fade — the model *turns off* surplus capacity (Bishop’s figure 10.6 phenomenon; an Occam effect: the marginal likelihood charges rent for parameters that explain nothing). Start with generous K , let VB prune — an alternative to the model-selection sweep of Exercise 8.10, with the honest caveat that the pruning threshold is secretly set by the Dirichlet concentration: priors are knobs, and Module 5’s registry wants them logged.

One inheritance to keep in mind from D §D.4: the factorization (8.5.1) is a reverse-KL approximation, so the variational posterior *understates* parameter uncertainty — read VB error bars as lower bounds on your ignorance.

Checkpoint exercise

Exercise 8.4 [Bishop Ex. 10.5: variational inference with a point mass is EM]. [★★★★ Core] *Let the hidden variables comprise latents $\mathbf{z}$ and parameters $\boldsymbol{\theta}$, with variational family $q(\mathbf{z}, \boldsymbol{\theta}) = q_{\mathbf{z}}(\mathbf{z}) q_{\boldsymbol{\theta}}(\boldsymbol{\theta})$ and $q_{\boldsymbol{\theta}}(\boldsymbol{\theta}) = \delta(\boldsymbol{\theta} - \boldsymbol{\theta}_0)$, $\boldsymbol{\theta}_0$ a free parameter. (a) Substitute into the ELBO and show optimizing $q_{\mathbf{z}}$ at fixed $\boldsymbol{\theta}_0$ is exactly the E-step (posterior at $\boldsymbol{\theta}_0$), while optimizing $\boldsymbol{\theta}_0$ at fixed $q_{\mathbf{z}}$ maximizes $\mathbb{E}_{q_{\mathbf{z}}}[\log p(\mathbf{X}, \mathbf{z}, \boldsymbol{\theta}_0)]$ — the M-step with a log-prior term, i.e. MAP-EM (Module 4 §4.5’s move appearing automatically). (b) Handle the subtlety the exercise is secretly about: the differential entropy of δ is $-\infty$. Show it enters the ELBO as an additive constant in $\boldsymbol{\theta}_0$ (so coordinate updates are unaffected), and state what this says about comparing ELBO values across models when point-mass factors are involved. (c) Conclude the unification in one sentence: EM, MAP-EM, and VB are one algorithm family indexed by how richly $q_{\boldsymbol{\theta}}$ is parameterized — and locate §8.5’s two payoffs as consequences of widening $q_{\boldsymbol{\theta}}$ beyond a point.*

8.6 From GMM to mixture of experts

The translation the whole chapter exists for. Reinstate the inputs:

	GMM (this chapter)	Mixture of experts (the thesis)
Models	density of x	conditional density of y given x
Latent prior	π_k constant	$\pi_k(x) = \text{softmax}_k(g(x))$ — the gate
Component	$\mathcal{N}(x \mu_k, \Sigma_k)$	$\mathcal{N}(y f_k(x), \sigma_k^2)$ — the expert
Likelihood type	generative (joint in x)	discriminative (conditional; Module 4 §4.7)
E-step	(8.3.1)	identical in form: (B.8.4)
M-step, weights	$\pi_k = N_k / N$	weighted softmax regression toward γ (M4, Ex. 4.5(b))
M-step, components	weighted Gaussian fits (8.3.2)	weighted regressions (M4, Ex. 4.9; IRLS §4.8)

Three remarks complete the translation; Exercise 8.5 derives all of it. First, the E-step is *unchanged*: responsibilities are Bayes’ rule whether the prior over z is constant or feature-conditioned — the formula carried since (B.8.4) was already the general case. Second, the M-step’s two halves are Module 4 problems *verbatim*: the gate’s update is cross-entropy toward soft labels (stationarity $\pi(\mathbf{x}_n) = \gamma_n$ on average — the $\pi - r$ gradient of (B.8.6) reaching zero), and a linear-Gaussian expert’s update is exactly the weighted least squares of Bishop Ex. 3.3 — the forward reference planted in Module 4 §4.8.2 now lands. Third, with *neural* gate and experts the M-steps have no closed forms; replacing each by a few gradient steps gives *generalized EM* (any M-improvement preserves monotonicity), and taking that to its limit — one simultaneous gradient step on everything — recovers plain SGD on the mixture NLL, the equivalence proven in D §D.8 (tight bound at the E-step optimum $\Rightarrow$ identical gradients). *EM and backprop are not rival training procedures for the corrected NEC; they are two step-size policies on one objective.* What EM contributes irreplaceably is the *vocabulary*: responsibilities, effective counts, collapse, pruning — the diagnostic language in which gate behavior becomes legible.

Forward-looking context, one paragraph as the topic list requests. The same ELBO machinery with *amortized* variational posteriors and reparameterized gradients (D §D.7) is the variational autoencoder; with variational posteriors over network *weights*, the Bayesian neural network. Neither is on the thesis’s critical path; both become available, nearly free, if a stochastic gate or uncertainty-aware experts ever earn their complexity — and Foundations E’s Gumbel machinery (next) is the missing sampling piece for the discrete case.

Checkpoint exercise

Exercise 8.5 [Syllabus: EM for the mixture of experts, assembled]. [★★★★ Core] *The conditional mixture $p(y | \mathbf{x}) = \sum_k \pi_k(\mathbf{x}) \mathcal{N}(y | f_k(\mathbf{x}), \sigma_k^2)$, gate $\pi_k(\mathbf{x}) = \text{softmax}_k(g_k(\mathbf{x}))$. (a) E-step: derive the responsibilities from Bayes’ rule and verify they coincide with the variational optimum q^* of the ELBO decomposition — this is the syllabus’s “verify against q^* ” item; cite (B.8.4) and D, Ex. D.3(c), and write the two-line identification. (b) M-step, gate: show the gate’s subproblem is $\max \sum_n \sum_k \gamma_{nk} \log \pi_k(\mathbf{x}_n)$ — weighted cross-entropy toward soft labels — whose per-sample logit gradient is $\pi_k(\mathbf{x}_n) - \gamma_{nk}$ (Module 4, Ex. 4.5(b)); conclude the M-step’s stationarity condition and its reading (“the gate learns to predict the responsibilities”). (c) M-step, experts: for linear experts $f_k(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}_k$, show the subproblem is exactly Bishop Ex. 3.3’s weighted least squares (Module 4, Ex. 4.9) with weights γ_{nk} / σ_k^2 , and write its closed form; derive also the σ_k^2 update (responsibility-weighted mean squared residual). (d) Monotonicity and the SGD identification: cite D, Ex. D.3(d) for monotone full EM; then show that replacing each M-step by one gradient step, and letting the E-step be implicit, yields exactly one SGD step on the mixture NLL — use the tightness-at- q^* identity of D §D.8 to equate the gradients. State the practical conclusion for the corrected NEC’s training loop in one sentence.*

8.7 Capstone: training the corrected NEC, as a theorem with an implementation

Assemble the chapter into the thesis’s training procedure, each line load-bearing and cited.

Objective. Per-sample mixture NLL (8.2.1)’s conditional twin, computed as $-\text{LSE}_k(\log \pi_k(\mathbf{x}) + \log \mathcal{N}(y \mid f_k(\mathbf{x}), \sigma_k^2))$ in the log domain with the (B.6.10) stabilization — never as a log of a sum of densities (Exercise 8.1’s float64 cliff).

Training. AdamW on the NLL (Foundations C), understood as generalized EM (§8.6): every gradient step implicitly recomputes responsibilities and nudges gate and experts toward their weighted M-step targets. Optionally, explicit EM epochs (freeze responsibilities, take several expert/gate steps) as a stabilization variant — worth one ablation line in the thesis, not more.

Initialization (§8.4, mode 3, operationalized): gate logits near zero — a near-uniform gate, entropy near $\log K$ (B §B.6) — so no expert starts dead; experts *diversified* (different seeds; or pre-trained briefly on volatility-sorted data slices to break symmetry deliberately and interpretably); σ_k warm-started frozen (Module 6 §6.3’s heteroskedastic caution).

Monitoring (the failure modes as dashboards): NLL (monotone only under full M-steps — do not panic at SGD noise; C §C.3); gate entropy per sample and per batch; expert utilization N_k/N with alarms near zero (collapse twin); responsibility sharpness over training (healthy specialization sharpens gradually — instant sharpness is collapse, permanent uniformity is a dead gate).

Reporting (§8.4, mode 2): align expert labels across folds by a declared statistic (e.g. ascending expert σ_k , or gate-correlation with VIX) *before* any per-regime table; log restarts, K values, and prior/floor settings in the Module 5 registry; if VB-style pruning is used to choose K , report the concentration parameter as the hyperparameter it is.

Checkpoint (from the syllabus). Closed book: write the MoE likelihood and state its exact relation to the GMM (the dictionary of §8.6); state EM as coordinate ascent on the ELBO and prove monotonicity (or cite D, Ex. D.3(d), and reproduce it); derive the GMM M-step closed forms (8.3.2) from scratch; apply the log-sum-exp trick to a mixture likelihood and say exactly what fails numerically without it; and articulate why the variational view matters for any Bayesian or stochastic-gate extension (priors kill singularities and prune components; reparameterization is the gradient path through sampling — D §D.7, Foundations E next). If all five are comfortable, the thesis’s core estimator is yours from first principles.

8.8 Exercises

Importance labels. Each exercise carries a priority badge for the NEC/MoE thesis track, reflecting how load-bearing it is — *not* how hard it is. [★★★ Core] mark the five the checkpoint demands closed-book: the log-domain mixture pipeline (8.1), the M-step from scratch (8.2), the three failure modes with their protocol clauses (8.3), the VB-contains-EM unification (8.4), and the MoE assembly that *is* the corrected NEC’s estimator (8.5). [★★ Worth it] : mixture moments (8.6), the ESL notation bridge (8.9), and choosing K (8.10) — the last feeding a pre-registered thesis decision. [★ Low priority] : the complete-data baseline (8.7, contained in 8.2 with hard labels) and the Bernoulli boundedness contrast (8.8).

Questions only, as established. **Core exercises ([★★★ Core]) sit inline in the chapter body as checkpoint exercises, at the end of the section they test;** this section collects the supplementary exercises only. De-dup ledger for this set: Bishop 10.1 $\equiv$ Exercise D.3(a), not re-assigned; the

syllabus’s EM-monotonicity and coordinate-ascent exercises are D.3(d) and D’s eq. (D.5.4), cited inside Exercise 8.5 rather than repeated; the LSE-trick derivation is (B.6.10), with only its mixture-specific consequences assigned; ESL 8.1/8.2 appear as the notation bridge they truly are.

On mixtures and their likelihoods (Sections 8.1–8.2)

Exercise 8.6 [Bishop Ex. 9.12, extended]. [★★ **Worth it**] For the mixture $p(\mathbf{x}) = \sum_k \pi_k p(\mathbf{x} | k)$ with component means $\boldsymbol{\mu}_k$ and covariances $\boldsymbol{\Sigma}_k$: (a) derive $\mathbb{E}[\mathbf{x}] = \sum_k \pi_k \boldsymbol{\mu}_k$ and $\text{Cov}(\mathbf{x}) = \sum_k \pi_k (\boldsymbol{\Sigma}_k + \boldsymbol{\mu}_k \boldsymbol{\mu}_k^\top) - \mathbb{E}[\mathbf{x}] \mathbb{E}[\mathbf{x}]^\top$ via the tower property and the law of total variance (B, Lemma B.2.2 and Prop. B.2.3); (b) specialize to centered Gaussian components with equal means: recover exactly the setup of Foundations B Exercise B.6, and restate its conclusion (scale mixtures are leptokurtic) as a property of (8.1.1); (c) one sentence: why does (b) make the GMM the minimal generative account of fat-tailed returns rather than an arbitrary flexible family?

Exercise 8.7 [Bishop Ex. 9.7: the complete-data baseline]. [★ **Low priority**] Maximize the complete-data log-likelihood (8.2.2) (labels z_{nk} known, one-hot) over $(\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma})$. Show the problem separates: each component’s $(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ is the plain Gaussian MLE (B, Ex. B.2) over its own points, and π_k is the fraction of points labeled k (Lagrange multiplier on the simplex — Module 4 §4.3). Conclude in one sentence why (8.3.2) is exactly this answer with hard labels softened to responsibilities.

On EM (Section 8.3)

Exercise 8.8 [Bishop Ex. 9.15, paired with the boundedness contrast]. [★ **Low priority**] A mixture of multivariate Bernoullis: components $p(\mathbf{x} | \boldsymbol{\mu}_k) = \prod_{i=1}^p \mu_{ki}^{x_i} (1 - \mu_{ki})^{1-x_i}$ on $\mathbf{x} \in \{0, 1\}^p$. (a) Maximize the expected complete-data log-likelihood over $\boldsymbol{\mu}_k$ and derive the M-step $\boldsymbol{\mu}_k = N_k^{-1} \sum_n \gamma_{nk} \mathbf{x}_n$ — the same weighted-mean template as (8.3.2), emission family swapped. (b) Prove the incomplete-data log-likelihood of this mixture is bounded above (each factor is a probability ≤ 1), so no collapse singularity exists — and state precisely which property of continuous densities reintroduces the pathology for the GMM (§8.4, mode 1). (c) One sentence: what does this contrast imply about mixtures over discretized return buckets as a robustness variant?

Bridges and unifications (Sections 8.5–8.6)

Exercise 8.9 [ESL Ex. 8.1–8.2, as the notation bridge they are]. [★★ **Worth it**] ESL §8.5 writes EM via $R(\theta', \theta) = \mathbb{E} [\log p(\mathbf{X}, \mathbf{Z} | \theta') | \mathbf{X}, \theta]$ and a maximization over distributions $\tilde{P}(\mathbf{Z})$. (a) Build the dictionary: identify ESL’s $\tilde{P}$, R , and its lower-bound function with this book’s q , $\mathbb{E}_q[\log p(\mathbf{X}, \mathbf{Z} | \theta)]$, and $\mathcal{L}(q, \theta)$, term by term. (b) ESL 8.1 asks: show $\mathbb{E}_q[\log r(Y) / q(Y)]$ is maximized over densities r at $r = q$. Recognize this as Gibbs’ inequality (B, Theorem B.6.1 / Ex. B.5) and give the one-line proof in ESL’s notation. (c) ESL 8.2 asks: show the optimal $\tilde{P}(\mathbf{Z})$ is the conditional $\Pr(\mathbf{Z} | \mathbf{X}, \theta')$. Recognize this as the Gibbs-maximizer result you proved as D, Ex. D.3(c), and translate that proof into ESL’s notation. (d) One sentence: why is owning this dictionary worth an exercise (what goes wrong when you read ESL §8.5 and Bishop Ch. 9 side by side without it)?

Exercise 8.10 [Supplementary: choosing K]. [★★ **Worth it**] (a) Count the free parameters of a p -dimensional, K -component GMM with full covariances: $d(K) = (K - 1) + Kp + Kp(p + 1)/2$ (justify each term; the symmetric-matrix count is Bishop Ex. 2.21’s formula, derivable in one line). (b) Write the AIC and BIC criteria (Module 5 §5.2) for the GMM as functions of the maximized log-likelihood and $d(K)$, and explain the one structural caveat: the regularity conditions behind these criteria are violated at mixture boundaries (a component with $\pi_k = 0$), so treat the comparison as heuristic ranking, not calibrated testing. (c) Describe the held-out alternative — per-observation out-of-sample NLL, read as compression (D §D.2) — and why Module 5’s purging discipline applies to this likelihood comparison exactly as to any backtest. (d) One paragraph: compare all three routes (information criteria, held-out NLL, VB pruning) for the thesis’s choice of expert count $K \in \{2, 3, 5\}$, and state which combination you would pre-register and why.

Source map and what comes next

Section	Primary source(s)	Feeds directly into
8.1 GMM, moments, identifiability	Bishop §9.2	regime hypothesis; reporting rules
8.2 Two likelihoods	Bishop §9.2–9.3; ESL §8.5	why EM exists
8.3 EM executed	Bishop §9.2–9.4 + D	the estimator itself
8.4 Failure modes	Bishop §9.2 + own	gate diagnostics; thesis protocol clauses
8.5 VB mixtures, structurally	Bishop §10.1–10.2	K selection; singularity cure
8.6 GMM $\rightarrow$ MoE	Bishop §14.5 spirit + B §B.8	the corrected NEC’s training
8.7 Training capstone	synthesis	thesis experiments, Module 10

Two modules remain in Part IV. Module 9 adds *time* to the latent labels — Markov persistence, the forward–backward recursions, Baum–Welch (which you will recognize instantly as EM with responsibilities computed by dynamic programming), and the Hamilton filter that is the thesis’s classical baseline; its centerpiece comparison, Markov gating versus feature-conditioned gating, was pre-framed in Module 4 §4.7 and D §D.8. Foundations E then supplies the Gumbel machinery for Module 10’s stochastic-routing theory. The mixture mathematics is now complete; what follows is its dynamics and its engineering.